

Thinking with a Feminist Political Ecology of Air-and-breathing-bodies

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DOI: 10.1177/1357034X19900526

journals.sagepub.com/home/bod



Irma Kinga Allen 

KTH Royal Institute of Technology

Abstract

Social theory has paid little attention to air, despite its centrality to bodily existence and air pollution being named the world's biggest public health crisis. Where attention to air is found, the body is largely absent. On the other hand, conceptualizing the body without life-sustaining breath fails to highlight breathing as the ongoing metabolic bodily act in which the materiality of human and more-than-human intermingle and transmute one another. Political ecology studies how unequal power structures and knowledge production reproduce human–environment relations, including a nascent focus on the body and air – but as separate issues. This article argues that a political ecology of air would productively fuse with a political ecology of the body to bring the visceral realm into intersectional analysis of air's contemporary materialities. A feminist political ecology situates explicitly air-and-breathing-bodies, their intimately posthuman, relational, elemental and corpomaterial intra-action, at the heart of such analysis.

Keywords

air, breath, breathing body, embodiment, feminist political ecology, posthuman

It goes without saying that air is the ‘implicit condition of bodily existence’ (Sloterdijk, 2009). But that *going without saying* is precisely why air and, more so, breath have received relatively little attention within Western social theory, including body studies. Air has largely been unaired; breath, moreover, barely spoken of. ‘Air is left to drift . . . neither theorized nor examined, taken simply as solidity’s lack’, muses Timothy Choy (2011: 9–10). Similarly, Nile Green

Corresponding author: Irma Kinga Allen. Emails: irma.allen@abe.kth.se; ikallen@kth.se

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(2008: 283–284) explains that breathing has withheld theoretical scrutiny appearing ‘neither to reflect nor require a context’ as ‘one of the last bastions of the universal’. Even where air and breath *are* theorized, they are usually discussed in separate breaths, as distinct categories, while breathing as a lived phenomenon is usually treated in the context of *ill*-health – as an absence or site of suffering.

Recently, however, we can begin to discern an uplift into air within Western academic research focused on human–environment relations. Within geography, there is an emerging ‘aerography’ (Jackson and Fannin, 2011) and literature on atmospheres (Anderson, 2009) and verticality (Graham, 2016). In anthropology, there is growing ‘suspension’ of thought (Choy and Zee, 2015). In literary studies, we find exploration of air’s cultural meanings and a turn to the elemental (Cohen and Duckert, 2015; Connor, 2013). These perspectives do enormous amounts to further our still tentative considerations of our airy selves and worlds. Yet there is ample room too to draw more attention to living bodies, human and more-than-human, within these diverse approaches. Bodies in intra-relationship with air often remain either conspicuously absent or treated as objects, or surfaces upon which technologies, discourses and practices act; ideas rather than felt presences. Specifically, breathing, as the practice that connects bodies to air, is under-theorized or overlooked, marking this Special Issue on breathing an important contribution. Body studies is poised to bring embodied perspectives on our ecological embeddedness in air through breath. This is particularly salient at a time when the World Health Organization has declared air pollution the world’s biggest public health crisis.

In this article, I want to think through a Feminist Political Ecology of air-and-breathing-bodies, which takes the relational condition of breath, bodies and airs as its starting point. I propose that an emerging political ecology of air is best approached through combining with a nascent Feminist Political Ecology of the body, using breath as a pivot to enrich situated perspectives on how air-and-breathing-bodies come to matter in uneven and differentiated ways. The *Feminist* in Political Ecology calls to attention intimate scales, the previously considered immaterial and invisible, the intersectional aspects of corporeality and the relational, emotional and affective dimensions of lived experience and embodiment of airs and breath. As such, this contribution positions itself within debates in this

journal that have considered what the unbounded, non-discrete, relational, indeterminate, affective body may mean for our understanding of corporeality (Blackman, 2010, 2016; Blackman and Venn, 2010; Erin Manning, 2010; Yusoff and Waterton, 2017). As Lisa Blackman (2010: 4) has stated: 'rather than existing as bounded, autonomous subjects, we coexist in shared ecologies'. While this journal has engaged in considering embodied weather experiences (Allen-Collinson et al., 2018), bodily experiences 'in' nature (Macnaghten and Urry, 2016) and the indeterminate condition of toxic embodiment (Yusoff, 2017), there is still necessity for a broader engagement with those shared ecologies, their implications for embodiment and particularly air-and-breath's role in their lived experience.

In the time of the so-called anthropocene, 'human bodies are subject to restless, mutable environments' that have become the 'primary driving forces for human development and bodily change over lifetimes' (Lock, 2018: 4), such as via epigenetics (Blackman, 2016). Since anthropocenic ecological effects are unequally distributed, indeterminate, yet newly normative (Yusoff and Waterton, 2017: 5), with profound effects on embodiment, both human and more-than-human (De Wolff, 2017), it seems a pertinent moment for body studies to engage with critical political ecology, and vice versa, as a framework for discerning and disentangling some of these politicized processes that have got 'under the skin'. While (ill)health (which has its own political ecology trajectory – see King, 2010) is central to such interrogation, including in relation to breath/breathlessness (see *Life of Breath*), it will not be the focus of this article. Rather, health/ill-health is considered a fragile continuum that is always-already contained within bodily relations, where deterioration, abrasion, de-composure and emergence are intrinsically part and parcel of staying alive.

Body studies can learn from Feminist Political Ecology of the body by engaging with how the overlapping forces of power, discourse and matter, including affect and emotion, are implicated in the political ecological materialization and enactment of the body and embodiment through air and breath. As Jessica Hayes-Conroy and Allison Hayes-Conroy (2013: 81) argue, political ecology offers a framework that can 'meaningfully and practically operationalize' scholarship on bodily matter, including the 'often ethereal, yet

deeply vital scholarship on affect and emotion'. By avoiding the pitfalls of a potential depoliticization that attention to affect (particularly split from meaning-making (Wetherell, 2015)) and post-human relational ontologies can pose, as some scholars have argued (Malm, 2018), through a flattening of power relations or abstraction to a totalizing non-representational realm, a Feminist Political Ecology of air-and-breathing-bodies can maintain the focus on 'simultaneously structural and post-structural, material and discursive' (Hayes-Conroy and Hayes-Conroy, 2013: 82) accounts. At the same time, Feminist Political Ecology of the body can learn from body studies' historically broader perspective on the ontological and epistemological implications of a revitalized corporeality.

This article offers itself as a review of an undisciplined range of literature on airs, atmospheres, elements and breath gathered together through a Feminist Political Ecological frame. It takes seriously Mark Jackson and Maria Fannin's (2011: 438) invitation with regards to 'aerographies' to 'let the conceptual and disciplinary landscape fall where it may. Begin not with [any one discipline] but with the stuff itself'. What happens if we tune into breath to engage with political ecologies of air? Where could this take Feminist Political Ecologies of air-and-breathing-bodies? Breathing is thus followed both as speculative mnemonic and method. Studying the mutable, indeterminate ecological body has its methodological complications. Kathryn Yusoff and Claire Waterton (2017: 6) in a Special Issue on Indeterminate Bodies make the case that sensibility to volatile and un-know-able (in the traditional sense) (de/re)materializations require 'novel forms of engagement' with 'forms of knowing that fall outside and in excess of the demands of empiricism and its metrics'. In a Special Issue on Affect, Lisa Blackman and Venn Couze (2010: 9) also propose 'other ways of noticing' than the 'technologies of observation, seeing and listening' which characterize research in the humanities. This requires methods that pay attention to 'that which remains in excess to the practices of the speaking subject'. I posit breathing, or paying attention to and following breath, as one kind of experimental method that can reveal subtle and subtending insights about the politics of our ecological embodied relations with air. Attending to breath can open research up to different intimate and insightful frames of thinking political ecologies anew.

To begin, I outline political ecology and its engagement with air, before tracing the emergence of Feminist Political Ecology and what is specifically *feminist* about air-and-breathing-bodies. I then lay out three ways that the immaterial becomes substantial through such a perspective drawing on a whirlwind of literature that corresponds with air's irreverence for discipline or boundaries. Firstly, the breathing body renders air and breath, long thought immaterial, *material* in subtle, elemental, ways with implications for politics of location. Secondly, it brings intersectional organ-isms and labouring in-dustrial bodies to the fore as explicit research subjects. Thirdly, the breathing body helps us to pinpoint the lived experience of the embodiment of air in everyday ways, including emotional, affective and atmospheric encounters that shape political ecological claims and subjectivities, but have largely been ignored.

Feminist Political Ecology of Air-and-breathing-bodies

Political ecology is an interdisciplinary field that examines how inequitable power relations and their discourses shape the way human–nature relations are framed, articulated, experienced and produced. It emerged in the 1970s out of a concern for a perceived *apolitical* ecology, which takes Nature as a given and explains environmental degradation and resource access issues through a techno-managerial lens. Paul Robbins (2012: 20) defines it as stressing ‘not only that ecological systems are political, but also that our very ideas about them are further delimited and directed through political and economic processes’. When it comes to air pollution research, *apolitical* medical, technical and environmental health literature predominates. Yet air has received far less attention within political ecological literature than its sister element water (see, however, Buzzelli, 2008; Harper, 2004; Mitman, 2007; Véron, 2006). In the 2015 International Handbook on Political Ecology, air is not even an index entry.

In response to this gap, Stephen Graham (2015) sketches out what a ‘political ecology of urban air’ might entail, outlining 10 directions for research. Graham is particularly interested in air’s technical manipulation and the inequalities apparent in its production and access, such as through the spatialization of pollution and air-conditioning, its construction and use, and, deploying Peter

Sloterdijk's (2009) concern with 'atmoterror', air's manipulations in warfare. Through this, Graham offers an important contribution to pushing a political ecology of air forward. What is missing, but implicit, however, is the living, breathing body. Air is still divorced from the experiences of human and more-than-human bodies that breathe it. This turns air into an object rather than an immersive presence or medium and overlooks much intimate, situated and *embodied* understanding of our relatedness to air.

Feminist Political Ecology emerged in the 1990s with an initial focus on bringing interrogation of the gendered dimension of the uneven production and experience of human–environment relations, including resource access and control, to light (Rocheleau et al., 1996). It developed to highlight the location of multiple scales of the political to include the community, household and the everyday (Elmhirst, 2015; Rocheleau, 2008), discursive concepts of culture, identity, subjectivity and situated knowledge and their interrelation with environmental conditions (Grove, 2009; Rocheleau, 2008; Sundberg, 2017), and, more recently, corpomateriality, or material embodiment, with focus on ecologies of 'bodies, emotions, everyday practices and relationships' (Harcourt and Nelson, 2015: 4).¹

Acknowledging the debt to feminist emphasis on the body, it is emerging as one of the key locusts for analysis and engagement for Feminist Political Ecology because of its grasp on lived experience. For example, Sapana Doshi (2016) makes five propositions for an urban political ecology of the body to include a focus on feminist-informed concerns with embodiment, social reproduction and political subjectivity. Further, drawing on feminist theory, Allison Hayes-Conroy and Jessica Hayes-Conroy (2015: 659) outline the most comprehensive framework for a 'Political Ecology of the Body', proposing the incorporation of the visceral realm – or 'the relational place where bodies feel' – into political ecological interrogation of 'wider economic structures and systems of meaning making' in human–environment matters. To do this, they propose a tripartite model that draws attention to the 'impact of structural forces, knowledge production, and visceral experience on the health and well-being of human (and more-than-human) bodies' (Hayes-Conroy and Hayes-Conroy, 2015: 662). In this way, they propose that political ecology's three phases of development should be brought together under one rubric: structural, discursive and new material.

It is the less-often included, and feminist-informed, third realm of corpomateriality that they emphasize would bring the impacts of material agency, affect and bodily experience into analysis of the reproduction of human–environment relations.

But what about the *breathing* body? Theorizing of breathing is not usual terrain for political ecology. One notable exception is Greg Mitman's (2007) *Breathing Spaces*, an environmental history of the role of allergy in the production of the American landscape. Yet, although the text is bookended by brief and poignant accounts of his personal experience of asthma, as with the health literature, the lived experience of breathing and of air, and their intertwined material meanings and politics, is not the main focus. Within Feminist Political Ecology, while there have been a number of studies of water access, use and distribution, as well as wider feminist works on bodies of water, the literature on air in this vein is also scant. Perhaps, the focus on water is because wetness, fluidity, leakiness and permeability have been part of feminist concerns with the female body since its early origins (Bird Rose, 2002; Bordo, 1995). Flow and fluidity have also correlated to thinking about the production of sexual difference (Canters and Jantzen, 2014; Grosz, 1994). But air and breath also reveal crucial aspects of permeability, relationality, vulnerability and also excessive un-tameability that speak to feminist rethinkings of the body as thoroughly open, unstable, fluxed and flexed – part of a processual ontology of the corporeal. For air is the matter we are most defenceless against; connected in its unseen tactility, we move and breathe, without realizing, through and with a whole airy world that touches, permeates and enfolds Self-and-Others in a shared soup. Dust, molecules, atoms, gases and water vapour, microbes, viruses and pollen pass into our lungs with each breath of air. Are these not the most intimate of matters?

But where academic theory has undergone a 'material turn', air and particularly breath have yet to be incorporated fully as non-human, agentic entities. Perhaps, this is due to an arguably masculinist 'logic of solids' on which our intellect and concepts have been formed (Henry Bergson in Jackson and Fannin, 2011: 435). Peter Adey (2014:82, 102) finds a gendering of the air in common with a longer cultural history of the senses in which the oppressed feminine is associated with bodily senses of smell and witchy airs. Likewise, airy-fairy, air-headed, ungrounded, groundless, gassy, cloudy,

windy, long-winded or breathy thinking-writing are considered negatives and often coded as feminine ('air-headed' calling to mind the stereotypical 'blonde'). As Jackson and Fannin (2011: 436–437) urge, however,

we need to address how our elemental assumptions, which are wrapped up within phenomenological pretexts, inflect what it is we take matter to mean . . . If we begin with the assumption that the elemental is the material as solid, then we limit drastically the scope and relevance of our conceptual ecologies.

Beyond an acknowledgement of relational ontology, feminism has often extended political analysis into supposedly immaterial and intimate domains such as the home, the body and the personal. Thus I propose that feminist analysis, particularly Feminist Political Ecology, is well positioned to retrieve air and breath as micro-candidates for materialization through bodies in our ecological present.

Indeed, critical of the foundation of philosophy on solidity, it was feminist philosopher, Luce Irigaray (1999) who argued that the logocentric metaphysics of Heidegger and his Eurocentric peers and heirs (not airs) is borne of a culture forgetful of life – one that neglects to include respect for air and breath in its consequently anaerobic concepts, systems of thought and practices, excluding sexual difference in the process. For Irigaray, air is the mediator of all perception, knowledge, thought, language, imagination and action. Thought without air is therefore unthought. Breathing, moreover, is the practice which connects us to an ontology of 'vegetal being' (Irigaray and Marder, 2016) – a bodily codependency on respiring plant life, skies and seasons. Thus, Irigaray points to how attention to breath, and so air, is a feminist re-ordering of the processual, relational unfolding of life.

Magdalena Górská (2016) has, more recently, also explicated a material and embodied perspective on the feminist vulnerability politics of breathing. For her, breath brings the feminist concerns of intersectionality and vulnerability into account by rendering the politics of breathing a matter of corpomaterial difference, post-human relatedness, agency and embodied situatedness. She draws on feminist theorist Karen Barad's notion of 'agential realism' to conceptualize the way that 'corporealities' (rather than 'bodies' as such, which may preclude attention to material agentiality and smother

over with a sense of universalizing and essentializing norms) occur as phenomenal emergent constitutions of material-discursive, 'intra-active' becomings of environments, cultures, technologies, affects, discourses, social power relations and so on. As Górska (2016: 47) states, quoting Barad: 'To rephrase . . . *Phenomena* (such as breathing), then, do not pre-exist materialization processes but come into being within intra-active dynamics as "differential patterns of mattering"'. Górska also draws on feminist Stacy Alaimo and Susan Hekman's (2008, 238) concept of 'trans-corporeality' to explicate her understanding of embodiments to be 'entangled territories of material and discursive, natural and cultural, biological and textual'. Bodies and environments are mutually constitutive as intersectional vulnerabilities, and breathing enacts this at the microscale. Thus, Górska's (2016: 34, emphasis added) pioneering work emphasizes the explicitly '*feminist* respiratory politics' at the centre of her thesis.

Irigaray considered a philosophy of breathing to be a 'feminine act of remembering'. Here, inspired by Górska, I suggest a political ecology of air-and-breathing-bodies as a *feminist* act of remembering. The hyphenations mark the always-already situatedness and embeddedness of bodies-in-air performed through the ongoing process of explicit breath, attending to the politics inscribed in air, and articulating the politics implied in air's material-discursive intra-action with human-more-than-human bodies. The emphasis on 'political' in this ecology ensures such analysis, while leaving the ground, never strays far from renderings of power that are immanent and emergent. Thus, by bringing together two nascent but currently separate areas – the political ecology of air and Feminist Political Ecology of bodies – through breath, useful accounts of air particularly as a lived immersion, its contested politics, discourses, meanings, perceptions and ever-relatedness to breath could enable us to better ascertain air-and-breath's *experienced* uneven distributions and claims. In turn, for body studies, this develops conceptions of an ecological body into new breathy–airy spaces.

In the remainder of this article, I will outline three lines of breath that Feminist Political Ecological thinking with air-and-breathing-bodies could take airy organ-isms, rendering material the elemental, lungful and affective. Breath acts as a feminist mnemonic that we are airy bodies/bodies of air (not only nor primarily of water – see Neimanis, 2017), nested within and collectively (re)producing air's

multiple, uneven body masses. The breathing body as feminist method – towards cultures not forgetful of life.

Breath #1: Elemental Materialism of Air, Breath, Bodies and Place

Having outlined how air-and-breath are overdue candidates for feminist re-theorization as material within political ecologies, one way of conceptualizing such a re-mattering is through considering air-and-breath as not necessarily composed of bits of matter (the shared ‘soup’ described earlier) but as elemental; elemental not in the sense of essence but in the sense of the chemical. Peter Adey (2015: 61), using air as a focal example, urges us to think of the elemental not as a *thing* requiring more powerful microscopes but as more-than – as force, attunement, background, condition, affinity and chemistry that exceeds thing-ontology and governance. Echoing Irigaray, for Adey the elements are anterior to perception and go beyond it, both force and force field. Thus, the elements should be thought of ‘not merely as substances but as gathering tendencies towards structures of feeling, such as practices, thoughts, ideas and sensations structured by the elemental’ (Adey, 2015: 61).

Taking off from the feminist philosophy of Irigaray, if we take bodies as primarily *elemental*, that is, composed of swirling mixtures and processes of earth, air, water and fire, bodies and air cannot fall out into discrete distinguishable entities but are fused through embodied breath. The connectivities between bodies-and-worlds are not so much individual but (al)chemical – collectively attuned or abused. Most recently, feminist theorist Astrida Neimanis (2017: 4) has posited water as the grounds for elemental embodiment, exploring the troubling challenge ‘bodies of water’ pose for the three main ‘pillars of humanist corporeality – discrete individualism, anthropocentrism and phallogocentrism’. Yet it is through air-and-breath that our elementally threaded together bodies are rendered radically open and connected, as much as through water, if not more perpetually. As Jeffrey Jerome Cohen and Lowell Duckert (2015: 7) argue, the elements are ‘partners in world-making’ and ‘are the outside that is already within’. Likewise, Tim Ingold (2010: S126) maintains that the ‘elemental forces’ of earth and sky, ground and air, are intrinsic to what mindful bodies *are*. Bodies in turn *inhabit*, rather than ‘live on’

or ‘in’ the simultaneously aerial and terrestrial world – and it is only through breathing that we are able to know that world. Thus, breathing is a form of knowing – proposed earlier as part of a feminist method of engagement with the corporeal–ecological nexus.

Taking this up, Choy (2011: 12) has done much thinking with air and breath simultaneously through his ethnographic work on air pollution in Hong Kong, where ‘[p]art of air’s substantiability . . . comes from the fact that it is always breathed’. He conceives of air as co-constituted with ‘breathers’, who experience air and the airborne as elements (Choy, 2011: 24). Air’s elemental materiality is thus announced through breath and the breathing body forges partnership with air’s elemental manifestations. Breath figures the impossibility of pinpointing the moment the human melts into air and vice versa. It is both human and becoming non-human, an aspect of the body but extending beyond it. Indeed, in *Caesar’s Last Breath*, Kean (2017) reminds us that historical bodies, ancient matter, are still adrift on the airwaves. Even 2000 years later, some of us are still breathing in the molecules of the last breath of Caesar, implying a deeper temporality to air, and thus to bodies, that is usually granted only to the geo-logic.

Breath itself can be thought of as elementally material too. While breathing refers to the physical process through which lungs draw in air, and respiration is the chemical process by which gases are exchanged, breath itself can be understood as the elemental substance produced by this perpetual cycle. I take ‘substance’ in the Baradian sense of matter not as ‘fixed essence’ but an ‘intra-active becoming—not a thing but a doing, a congealing of agency’ (quoted in Alaimo and Hekman, 2008: 139). In this way, air-and-breathing-bodies collaboratively and continually make and remake one another and the world we inhabit through breath as its own intra-acting agentic force.

Adey poses the question – does this flight into a quasi-alchemical elementalism signal a break from the political? He believes not, since ‘particular affinities and elements make and remake certain kinds of conditions that can be subjected to critique’ (Adey, 2015: 60). Science, art and industry (and I would add ecology) all have particular affinities with the elemental as a ‘background or condition’. This is both representational and non-representational or material-discursive in inseparable emergence. The imaginary and symbolic

role that air plays in our lives also contains and pertains to material effects. This reverses Marx's famous dictum – in that all that is air, including our supposedly immaterial thoughts, melts into solids.

As such, a Feminist Political Ecology of air-and-breathing-bodies, imbued by an elemental materialism, would seek to unearth and give voice to the way air informs and structures those conditions, through channels, layers and linkages; the practices, thoughts, ideas and sensations of which Adey speaks. Or, how the elemental attunes and exerts pressure on the 'politics of location' – that geography of 'where we are', starting with the body (Rich quoted in Neimanis, 2013: 24). This must include the very breath that animates and situates that politics of location. If a 'living, breathing body is at once a body-on-the-ground and a body-in- the-air' (Ingold, 2010: S122), and the body is the site of the located political, then the implication is the shifting of the very space that counts *as* the political.

Indeed, Marijn Nieuwenhuis (2016b: 303) argues for an 'elemental ontology' that resituates a notion of relationally emergent place to include 'surroundings historically unrecognized as such' – including air and sea. His aim is to depart from an often racialized, bounded territoriality, the 'myth of a stable land', that underpins our received notions of the political as being 'rooted in the earth', the central notion of the idea of *geo-politics* (Nieuwenhuis, 2016b: 310, 313). Instead, Nieuwenhuis (2016b: 314) seeks to grant 'equal relational and platial powers' to earth, air and water. He gives credence to the breathing body specifically as the key locus through which this elemental power is expressed and experienced in the production of relational place (Nieuwenhuis, 2016a). This shifts our understanding of where the political resides – not on the ground, as common parlance would have it, but within *elemental place*. The production of place, as well as space, is lived and experienced within politicized relations of power that operate three-dimensionally, voluminously (see Bille, 2017; Bridge, 2013; Elden, 2013) or, simply – elementally. Alaimo's (2010b) concept of trans-corporeality also grounds the corporeal within an extensive posthuman *place*, emphasizing the vulnerable intra-action between elemental human bodies and non-human creatures and physical landscapes as the true site of both ethical and political space. The implication for a Feminist Political Ecology through air-and-breathing-bodies is a repositioning of the human into a broader political–elemental grounding.

Breath #2: Intersectional Lungs and In-dust-rial labouring Organ-isms

‘Why should our bodies end at our skin?’, Donna Haraway asked 25 years ago when proposing cyborgian ontology (quoted in Murphy, 2013). This Journal and body studies as a whole have long questioned the simple reification of the body as a single bounded subject, emphasizing its relational, open and processual characteristics as ‘always more-than-one’ (Blackman and Venn, 2010: 8; Erin Manning, 2010), not least in a recent Special Issue on skin itself (Lafrance, 2018). Most forthrightly, a Feminist Political Ecology of air-and-breathing-bodies persistently reminds us that the boundaries between the inside and outside of embodied selves are under a process of constant negotiation between unruly, uncontainable substances and more-than-human others, mediated by, for example, tissue, blood, neurons and organs, as well as air, dust and microbes through breath. Thus, feminist engagement with air-and-breathing-bodies soon dives us into the inner, invisible bodily medley, rendering this, too, material.

New materialist feminists have called for a reintroduction of the tricky terrain of biology into feminist theorizing about agency. Samantha Frost (2011: 70) highlights Barad’s call for attention to how ‘anatomy and physiology’ matter to processes of materialization, while Anne Fausto-Sterling (quoted in Frost, 2011: 69) invites feminists to ‘accept the body as simultaneously composed of genes, . . . hormones, cells, and organs . . . and of culture and history’. Through the provocative imagery and embodied reality of a body-with-more-than-lungs in constant porous motion, the breathing body acts as a mnemonic, reminding us to pay attention to our hidden interior, what Vicki Kirby (2015: 501) calls, both ‘meat’ and ‘mind’, the ‘goop and spill of corporeal interiority, the bone, muscle and sinewy connections, the colons, tracts, membranous pouches and bags of provisional containment, the greens, reds, yellows and browns that pulse and ooze just under our skin’. Reviewing Elizabeth Wilson’s (2015) book *Gut Feminism*, Kirby (2015: 501) reminds us that our messy ‘internal, corporeal landscapes’ are ‘thoroughly political’ and must be accounted for.

Yet too often political ecological engagement with the body has stopped at its ‘presumed boundaries’ (Guthman and Mansfield,

2015: 558). But *Feminist Political Ecology* has done much to spotlight that ‘bodily ecologies cannot be treated as separate from environmental flows’ (Guthman and Mansfield, 2015: 559). Explicit attention to air-and-breathing-bodies prompts consideration of how bodily enmeshed organs, not just the body ‘as a black box’, are shaped by power, which in turn shapes exposure to (polluted) airs, discourse around exposure and responses. In this way, an organ-ism perspective deepens understanding of ‘how and why the physical, visceral body comes to desire, revolt or remain apathetic’ (Hayes-Conroy and Hayes-Conroy, 2015: 661) in the face of, for example, toxic air. ‘All air is not equal’ (Godfrey and Torres, 2016: 9), but neither are lungs to breath it with.

Lungs are said to differ across gendered, racial and ethnic categories (Braun, 2014; Carey et al., 2007). Such discourses shape ideologies of breathing, medical interventions and perceptions of health risks. Thus, Górska (2016) argues that breathing is not a homogenous phenomenon but is structured along intersectional – that is class, gender, race, and also corpomaterial – lines. Lung specificities, such as capacity and size, age, lifestyle, privilege and access to technologies, such as respiratory aids, can all shape the ways in which breathing occurs. In turn, the political agency of resultant matter (of lungs, breath, body) is co-constituted intersectionally (Górska, 2016: 96). Where you breathe matters; who breathes counts.

Górska (2016) and Phoebe Godfrey and Denise Torres (2016) argue that intersectionality operates beyond social categories and should be extended along material, non-human, agentic lines, where *ecological* distributed unevenness intra-acts with breathing bodies to co-produce one another. The uneven spread of the elemental materiality of air becomes understood as its own intersectional process that works on and through other social categories to reify or reproduce them in intra-action. Rebecca Elmhirst (2015: 524) situates this extension of intersectionality to include the role of nature in producing social differences as central to Feminist Political Ecology, while Julie Guthman and Becky Mansfield (2015: 560, 558) also underscore the politics of ‘environmentally produced bodily difference’ as critical to Feminist Political Ecological engagement. One such example of materially intersectionally produced organ-ic difference is

Alaimo's (2010a) 'proletarian lung', shaped by exposure to micro-particles. It brings us to the crucial matter of dust.

Dust, the smallest matter that the human eye can see, is an immediate corollary to breathing bodies; we cannot help but inhale vast quantities daily. Yet, like air and breath, dust has been declared a 'non-thing' that 'we rarely count among things that matter' (Parikka, 2013). For Joseph Amato (2001), however, 'dust impinges on human history'. It is also 'unbearably light garbage' that contributes to *making* us (Marder, 2016: 3). This is so particularly for industrial workers, for whom dust is a daily hazard that constructs class, gender and racial difference. Feminist Political Ecology of air-and-breathing-bodies thus takes us through the transcorporeal proletarian lung into the realms of embodied global labour and its relation to dust.

Jussi Parikka (2015) analyses factory workers in China and mine labourers in Congo in the transnational media technology production line, for whom the non-human particles of dust will not leave the tired, labouring, expendable human body, instead forming toxic expressions of inequitable relatedness. Górska (2016) reflects on dusty lungs and spitting black in an interview with a Czech coal miner. Graham (2015) thinks particularly about construction workers suffocating in the extreme heat of Dubai to build air-conditioned elite luxury flats. Thus, Parikka (2012: 1) states that:

It is already on the level of particles such as dust . . . that we encounter political dilemmas of inclusion and exclusion, exposure and security. Dust and air pollution in general are silent, violent aggressors that demonstrate the political urgency of the atmospheric condition.

Parikka (2013) goes as far as to declare the 21st century as, among other deteriorating things, 'the century of dust', exposing the 'residues in our modes of production'.

Industrial labour and the body at work have not been central concerns of neither body studies nor political ecology (Barca, 2015: 78; Wolkowitz, 2006). But for Stefania Barca (2014), work is the fundamental human–nature relation that metabolizes environments and bodies. **A class-based environmentalist prejudice which views industry and nature as oppositional is one reason why perhaps workers' struggles to defend their own labouring bodies have not been a core focus of political ecology.** Another is the erosion of labour power

under neoliberal capitalism (Barca, 2015). In unearthing an environmental history of working-class environmentalisms (Barca, 2012) and of work, Barca (2014: 9) calls for workers' bodies to be understood as 'meta-texts where the political ecology of industrial societies' has been 'written'. Industrial toxins in the air, water, biogeochemical cycles and dust meet workers' bodies first. A Feminist Political Ecology following air-and-breathing-bodies renders labour and dust of central concern for ecological enquiry and its intersectional mattering.

Breath #3: A Poetics of Breathing Bodies: The Lived Experience of Air

According to Sasha Engelmann (2015: 430, 431), not enough attention has been given to 'the affective and aesthetic dimensions of being in and witnessing air and atmosphere'. Following Choy, she calls this the 'poetics of air'. Reframing this as a 'poetics of air-and-breathing-bodies', we come back into our material selves: breathing becomes the very being and witnessing of these qualities and experiences. A Feminist Political Ecology of air-and-breathing-bodies can push the boundaries of studies of such immaterialities – emotion, affect and atmospheres – in relation to our ecological being and knowing of airs. For embodied knowledge encompasses all these aspects and, as Megan Wainwright (2017) suggests, breathing often constitutes the means by which environments come to be embodiedly known.

For Leila Harris (2015: 172), the emotional and affective are 'nascent' interests in Feminist Political Ecology that represent a 'vast and relatively untapped promise'. The Hayes-Conroys (2013: 185) argue that incorporating these immaterialities into analysis benefits discussions by prompting investigation of how 'the particular physicalities of specific bodies and specific embodied emotions mediate conflicts over resource access/use, challenges of environmental governance, and our humanenvironment relationships and struggles most broadly'. A growing number of feminist political ecological scholars challenge 'masculinist' conceptions of policy-relevance and theorize memory, identity, values, claim-making and resource use and management as thoroughly emotional, affective issues (Dallman et al., 2013; Doshi, 2016; Elmhirst, 2011; Harris, 2015; Sultana, 2011). I argue that paying attention to breathing-in-air as a feminist method,

or 'way of noticing', brings emotions and affect to the heart of analysis because breathing, emotions and affects are deeply entwined.

Margot Lyon (1994) specifically uses the example of the respiratory system to explore the co-constitutive relationship of body, society and emotion. She argues that if society can be understood as 'bodies in structured relationship with one another', and 'emotion inheres in and is a product of those relations', then respiration, which is deeply implicated in the generation and experience of emotion and vice versa through physiological as well as psychological and phenomenological processes, is 'implicated in the very basis of social organization' and 'part of the genesis of social order' (Lyon, 1994: 84). Altering respiratory rhythms and qualities are part of the ontology of emotion and thus fundamental in communicating subjective and physiological states. These in turn form part of our habitus that enacts social life, which also has rhythmic properties supported by respiratory functions. Lyon argues that this co-rhythmic function of respiration in group relations must also have 'affective consequences (and vice versa)', for respiration plays a key role in shaping the 'affective order' or the 'continual modulation in affect in ongoing social relations' mediated by emotion (Lyon, 1994: 100–101). Illustrating this intimate connection between breath, affect and emotion most vividly, Brian Massumi (quoted in Latham and McCormack, 2004: 706), uses this example to explicate affect: 'Walking down a dark street at night in a dangerous part of town, your lungs throw a spasm before you consciously see and can recognize as human the shadow thrown across your path'. Lungs, then, are highly emotionally and affectively attuned and paying attention to breath often reveals such invisible processes going on beneath the surface.

The Hayes-Conroys (2008: 462) relate affect to questions of post-human agency and materiality: affect is about how bodies come to be affected and this 'occurs in the visceral realm . . . in which the whole molecular ensemble of the minded body feels the world'. In a *Body & Society* Special Issue on Affect, Blackman and Venn (2010: 9) also place the 'turn to affect' in a post-human, relational rethinking of bodies as 'always thoroughly entangled processes' where affect is a 'particular kind of process-in-practice' that 'remains in excess of the practices of the 'speaking subject'. Margaret Wetherell (2015: 2) further defines affect as 'the ways in which bodies are pushed and pulled

in contemporary social formations'. The link between affect and emotion is complex and much debated. Lisa Blackman and John Cromby (2007: 6) state that affect can traverse bodies but tends to undo 'the notion of a singular or sovereign subject. Emotions, in contrast, are those patterned brain/body responses that are culturally recognisable and provide some unity, stability and coherence to the felt dimensions of our relational encounters'. Deborah Thien (2005: 451) offers an appealingly simple explanation of their interrelationship: 'Affect is the *how* of emotion' – used to describe how emotion moves. Both emotion and affect, then, are perceptible as embodied encounters if not as always consciously voiced experiences. Putting this into conversation with contemporary body–environment conversations, we can also begin to treat emotion and affect as co-constituted through our political ecological intra-action situated in breathing.

Sara Ahmed's (2004: 117) idea of 'affective economies' is useful here in which she treats affect as an economy where it does not *reside* in either subject or object but through its circulation, which creates 'the very effect of the surfaces or boundaries of bodies and worlds', including collective bodies such as of the nation. Thinking affective ecologies, rather than economies, points to a material conception of how the environment interplays with this circulation. For example, humans and plants have been found to leave moods 'hanging in the air' through biochemical and semiochemical information that are perceptible through olfactory experience but not consciously processed (Haviland-Jones et al., 2013), while air pollution has even been suggested to have adverse affects on not only mood and feeling but interpersonal attraction (Rotton et al., 1978). More recent studies have drawn links between air pollution and depression (Tait, 2017), making material the 'black cloud' associated with the condition, and even suicide (Ng et al., 2016). While we should not take these biological formulations at face value, they point to interesting areas of further enquiry regarding political ecologies of the affective constitution of air-and-breathing-bodies. In turn, the qualities of one's breath, shaped by air in intra-action, can mould one's relation to the socionatural world and produce different senses of self, morally, socially, politically and culturally. Air's environmental subjectivities, its affectivities and emotional experiences, are those of the breath.

Breath also connects us to broader atmospherics. Anderson (2009) has developed the concept of ‘affective atmospheres’ with a view to unsettle the distinction that is usually made between non-narrative and asignifying affect and narrative and semiotic emotion. ‘Atmospheres’ on the other hand ‘are singular affective qualities that emanate from but exceed the assembling of bodies’ and hold dualisms such as ‘presence and absence, materiality and ideality, definite and indefinite, singularity and generality – in a relation of tension’ (Anderson, 2009: 77). Kathleen Stewart (2011: 445) writes that atmospheres and their ‘attunements’ have ‘rhythms, valences, moods, sensations, tempos, and lifespans’. In particular, it seems, things ‘hang’ or ‘change’ ‘in the air’; emotions as atmospheric intensities are ‘breathed in’. Mikkel Bille et al. (2015: 31–33) make the case that atmospheres are ‘the sensuous and emotional feel of a place’ and are ‘always located in between experiences and environments’ thus are ‘materially ambiguous’. Breathing bodies are their barometer.

Toxic embodiment in particular requires ‘atmospheric attunement’ (Stewart, 2011). An emerging anthropological strand, ‘chemo-ethnography’ (Shapiro and Kirksey, 2017), specifically contextualizes atmospheres and their bodily attunements within material ecological, chemical–industrial relations. For example, Nicholas Shapiro (2015) writes that ‘affective processes of attending to the minute aberrations of the body and atmosphere are the primary means of discerning protracted and low-level encounters with domestic chemicals’. A Feminist Political Ecology engaged with air-and-breathing-bodies would, in this sense, seek to understand the conjoining of air, breath and atmospheres in relation to material environments. The breathing body performs moments of atmospheric becoming, attunement and sensing within an eco-social life, including what Shapiro (2015) calls the ‘chemical sublime’ – disquieting embodied apprehension of toxic harms.

Last Breath

This article has made the case for more elementally situated perspectives on air through following breath and staying close to the body. It has attempted to track and trace a specifically *Feminist* Political Ecological engagement with air-and-breathing-bodies, proposing that feminist theory, with its concern for porosity and vulnerability, is poised to open up embodied accounts of air, where air, breath and

bodies are taken as inseparable, to inform more nuanced and intimate political–ecological accounts of how they come to matter in intersectional intra-relationship. Breathing is proposed as the locus point for such an engagement. Through three breaths, breathing is posited as mnemonic and method that attunes investigation of airy embodiment into new elemental terrains. Attending to breath brings previously considered immaterialities (elements, lungs, dust, emotions, affects, atmospheres and breath itself) into sharp focus with implications for how environmental subjectivities and politics come into being and how embodiment figures through these encounters. Firstly, thinking bodies and politics of location as primarily elemental shifts the bodily grounds of the political into a broader post-humanly attuned, chemical spatiality. Secondly, the inner body and its labouring organs exposed to dust become key sites for embodied enquiry in relation to air's uneven distribution and effects. Thirdly, more fine-grained accounts of the lived experience of air are made possible via micro-sensory attending to emotion, affect and atmospheres through breath as the portal. Paying attention to breath is thus a means to engage with the more subtle, intimate, microscale, relational, corporeal and processual aspects of embodied ecologies; aspects that feminist theory has been at the forefront of claiming as political. Such a Feminist Political Ecology of air-and-breathing-bodies could not only bring about deeper understanding of when and how air's socio-natures are contested but also how and why they are embodiedly normalized and accepted, resigned to or endured (Gould, 1993).

ORCID iD

Irma Kinga Allen  <https://orcid.org/0000-0001-8199-151X>

Note

1. For more detailed accounts of the developments of Feminist Political Ecology, see Elmhirst (2015) and Sundberg (2017).

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Irma Kinga Allen is an ENHANCE ITN Marie Curie PhD Fellow at the Environmental Humanities Laboratory at the Division of History of Science, Technology and Environment at KTH Royal Institute of Technology, Sweden. Current postal address: 42 Hawthorn Way, Cambridge, CB4 1AX, UK.

This article is part of the *Body & Society* special issue on ‘Interdisciplinary Perspectives on Breath, Body and World’, edited by Andrew Russell and Rebecca Oxley.